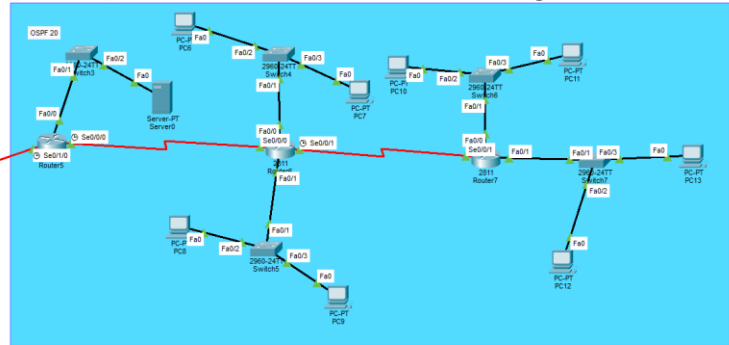
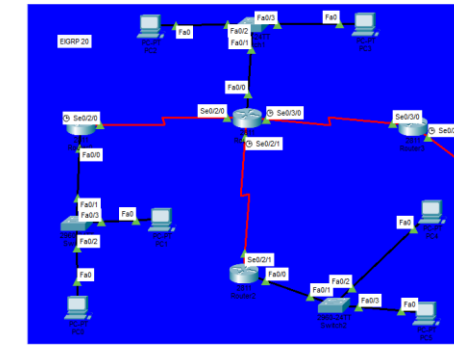


IP routes of all routers

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```

Codes: L - Local, C - Connected, S - Static, R - RIP, M - Mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

```

Gateway of last resort is not set

```

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C    10.1.1.0/30 is directly connected, Serial0/2/0
L    10.1.1.1/32 is directly connected, Serial0/2/0
D    10.1.2.0/30 [90/2681856] via 10.1.1.2, 00:03:28, Serial0/2/0
D    10.1.3.0/30 [90/2681856] via 10.1.1.2, 00:03:28, Serial0/2/0
D    10.1.4.0/30 [90/3193856] via 10.1.1.2, 00:03:28, Serial0/2/0
20.0.0.0/30 is subnetted, 3 subnets
D EX 20.1.1.0/30 [170/3219456] via 10.1.1.2, 00:03:28, Serial0/2/0
D EX 20.1.2.0/30 [170/3219456] via 10.1.1.2, 00:03:28, Serial0/2/0
D EX 20.1.3.0/30 [170/3219456] via 10.1.1.2, 00:03:28, Serial0/2/0
172.16.0.0/16 is variably subnetted, 13 subnets, 2 masks
C    172.16.10.0/24 is directly connected, FastEthernet0/0
L    172.16.10.1/32 is directly connected, FastEthernet0/0
D    172.16.11.0/24 [90/2172416] via 10.1.1.2, 00:03:28, Serial0/2/0
D    172.16.12.0/24 [90/2684416] via 10.1.1.2, 00:03:28, Serial0/2/0
D EX 172.16.20.0/24 [170/3219456] via 10.1.1.2, 00:03:28, Serial0/2/0
D EX 172.16.21.0/24 [170/3219456] via 10.1.1.2, 00:03:28, Serial0/2/0
D EX 172.16.22.0/24 [170/3219456] via 10.1.1.2, 00:03:28, Serial0/2/0
D EX 172.16.23.0/24 [170/3219456] via 10.1.1.2, 00:03:28, Serial0/2/0
D EX 172.16.24.0/24 [170/3219456] via 10.1.1.2, 00:03:28, Serial0/2/0
D EX 172.16.30.0/24 [170/3219456] via 10.1.1.2, 00:03:27, Serial0/2/0
D EX 172.16.31.0/24 [170/3219456] via 10.1.1.2, 00:03:27, Serial0/2/0
D EX 172.16.32.0/24 [170/3219456] via 10.1.1.2, 00:03:27, Serial0/2/0
D EX 172.16.33.0/24 [170/3219456] via 10.1.1.2, 00:03:27, Serial0/2/0

```

```

Hammadullah_R0#
Hammadullah_R0#
Hammadullah_R0#
Hammadullah_R0#

```

Copy

Paste

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Hammadullah_R1>en
Hammadullah_R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

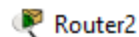
Gateway of last resort is not set

```
    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/2/0
L       10.1.1.2/32 is directly connected, Serial0/2/0
C       10.1.2.0/30 is directly connected, Serial0/3/0
L       10.1.2.1/32 is directly connected, Serial0/3/0
C       10.1.3.0/30 is directly connected, Serial0/2/1
L       10.1.3.1/32 is directly connected, Serial0/2/1
D       10.1.4.0/30 [90/2681856] via 10.1.2.2, 00:04:05, Serial0/3/0
    20.0.0.0/30 is subnetted, 3 subnets
D EX    20.1.1.0/30 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
D EX    20.1.2.0/30 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
D EX    20.1.3.0/30 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
    172.16.0.0/16 is variably subnetted, 13 subnets, 2 masks
D       172.16.10.0/24 [90/2172416] via 10.1.1.1, 00:04:05, Serial0/2/0
C       172.16.11.0/24 is directly connected, FastEthernet0/0
L       172.16.11.1/32 is directly connected, FastEthernet0/0
D       172.16.12.0/24 [90/2172416] via 10.1.3.2, 00:04:05, Serial0/2/1
D EX    172.16.20.0/24 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
D EX    172.16.21.0/24 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
D EX    172.16.22.0/24 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
D EX    172.16.23.0/24 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
D EX    172.16.24.0/24 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
D EX    172.16.30.0/24 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
D EX    172.16.31.0/24 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
D EX    172.16.32.0/24 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
D EX    172.16.33.0/24 [170/2707456] via 10.1.2.2, 00:04:05, Serial0/3/0
```

```
Hammadullah_R1#|
```

Copy

Paste

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

Hammadullah_R2>en

Hammadullah_R2#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks

D 10.1.1.0/30 [90/2681856] via 10.1.3.1, 00:04:37, Serial0/2/1

D 10.1.2.0/30 [90/2681856] via 10.1.3.1, 00:04:36, Serial0/2/1

C 10.1.3.0/30 is directly connected, Serial0/2/1

L 10.1.3.2/32 is directly connected, Serial0/2/1

D 10.1.4.0/30 [90/3193856] via 10.1.3.1, 00:04:36, Serial0/2/1

20.0.0.0/30 is subnetted, 3 subnets

D EX 20.1.1.0/30 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

D EX 20.1.2.0/30 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

D EX 20.1.3.0/30 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

172.16.0.0/16 is variably subnetted, 13 subnets, 2 masks

D 172.16.10.0/24 [90/2684416] via 10.1.3.1, 00:04:36, Serial0/2/1

D 172.16.11.0/24 [90/2172416] via 10.1.3.1, 00:04:37, Serial0/2/1

C 172.16.12.0/24 is directly connected, FastEthernet0/0

L 172.16.12.1/32 is directly connected, FastEthernet0/0

D EX 172.16.20.0/24 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

D EX 172.16.21.0/24 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

D EX 172.16.22.0/24 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

D EX 172.16.23.0/24 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

D EX 172.16.24.0/24 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

D EX 172.16.30.0/24 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

D EX 172.16.31.0/24 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

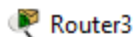
D EX 172.16.32.0/24 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

D EX 172.16.33.0/24 [170/3219456] via 10.1.3.1, 00:04:36, Serial0/2/1

Hammadullah_R2#

Copy

Paste



Physical Config **CLI** Attributes

IOS Command Line Interface

%DUAL-5-NBRCHANGE: IP-EIGRP 20: Neighbor 10.1.2.1 (Serial0/3/0) is up: new adjacency

Hammadullah_R3>en

Hammadullah_R3#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
 i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
 * - candidate default, U - per-user static route, o - ODR
 P - periodic downloaded static route

Gateway of last resort is 10.1.4.2 to network 0.0.0.0

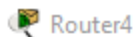
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks

D 10.1.1.0/30 [90/2681856] via 10.1.2.1, 00:05:14, Serial0/3/0
 C 10.1.2.0/30 is directly connected, Serial0/3/0
 L 10.1.2.2/32 is directly connected, Serial0/3/0
 D 10.1.3.0/30 [90/2681856] via 10.1.2.1, 00:05:14, Serial0/3/0
 C 10.1.4.0/30 is directly connected, Serial0/2/0
 L 10.1.4.1/32 is directly connected, Serial0/2/0
 20.0.0.0/30 is subnetted, 3 subnets
 D EX 20.1.1.0/30 [170/2195456] via 10.1.4.2, 00:05:14, Serial0/2/0
 D EX 20.1.2.0/30 [170/2195456] via 10.1.4.2, 00:05:14, Serial0/2/0
 D EX 20.1.3.0/30 [170/2195456] via 10.1.4.2, 00:05:14, Serial0/2/0
 172.16.0.0/24 is subnetted, 12 subnets
 D 172.16.10.0/24 [90/2684416] via 10.1.2.1, 00:05:14, Serial0/3/0
 D 172.16.11.0/24 [90/2172416] via 10.1.2.1, 00:05:14, Serial0/3/0
 D 172.16.12.0/24 [90/2684416] via 10.1.2.1, 00:05:14, Serial0/3/0
 D EX 172.16.20.0/24 [170/2195456] via 10.1.4.2, 00:05:14, Serial0/2/0
 D EX 172.16.21.0/24 [170/2195456] via 10.1.4.2, 00:05:14, Serial0/2/0
 D EX 172.16.22.0/24 [170/2195456] via 10.1.4.2, 00:05:14, Serial0/2/0
 D EX 172.16.23.0/24 [170/2195456] via 10.1.4.2, 00:05:14, Serial0/2/0
 D EX 172.16.24.0/24 [170/2195456] via 10.1.4.2, 00:05:14, Serial0/2/0
 D EX 172.16.30.0/24 [170/2195456] via 10.1.4.2, 00:05:13, Serial0/2/0
 D EX 172.16.31.0/24 [170/2195456] via 10.1.4.2, 00:05:13, Serial0/2/0
 D EX 172.16.32.0/24 [170/2195456] via 10.1.4.2, 00:05:13, Serial0/2/0
 D EX 172.16.33.0/24 [170/2195456] via 10.1.4.2, 00:05:13, Serial0/2/0
 S* 0.0.0.0/0 [1/0] via 10.1.4.2

Hammadullah_R3#

Copy

Paste

Physical Config **CLI** Attributes

IOS Command Line Interface

FULL, Loading Done

Hammadullah_Core#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 4 subnets
D 10.1.1.0 [90/3193856] via 10.1.4.1, 00:05:38, Serial0/2/0
D 10.1.2.0 [90/2681856] via 10.1.4.1, 00:05:38, Serial0/2/0
D 10.1.3.0 [90/3193856] via 10.1.4.1, 00:05:38, Serial0/2/0
C 10.1.4.0 is directly connected, Serial0/2/0
20.0.0.0/30 is subnetted, 3 subnets
C 20.1.1.0 is directly connected, Serial0/3/1
O 20.1.2.0 [110/128] via 20.1.1.2, 00:26:42, Serial0/3/1
O 20.1.3.0 [110/192] via 20.1.1.2, 00:26:42, Serial0/3/1
30.0.0.0/30 is subnetted, 1 subnets
C 30.1.1.0 is directly connected, Serial0/2/1
172.16.0.0/24 is subnetted, 12 subnets
D 172.16.10.0 [90/3196416] via 10.1.4.1, 00:05:38, Serial0/2/0
D 172.16.11.0 [90/2684416] via 10.1.4.1, 00:05:38, Serial0/2/0
D 172.16.12.0 [90/3196416] via 10.1.4.1, 00:05:38, Serial0/2/0
O 172.16.20.0 [110/65] via 20.1.1.2, 00:24:37, Serial0/3/1
O 172.16.21.0 [110/129] via 20.1.1.2, 00:26:42, Serial0/3/1
O 172.16.22.0 [110/193] via 20.1.1.2, 00:26:42, Serial0/3/1
O 172.16.23.0 [110/129] via 20.1.1.2, 00:26:42, Serial0/3/1
O 172.16.24.0 [110/193] via 20.1.1.2, 00:26:42, Serial0/3/1
S 172.16.30.0 [1/0] via 30.1.1.2
S 172.16.31.0 [1/0] via 30.1.1.2
S 172.16.32.0 [1/0] via 30.1.1.2
S 172.16.33.0 [1/0] via 30.1.1.2

Hammadullah_Core#

Copy

Paste

Router5

Physical
Config
CLI
Attributes

IOS Command Line Interface

```

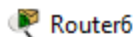
Hamadullah_R5>en
Hamadullah_R5#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 20.1.1.1 to network 0.0.0.0

 10.0.0.0/30 is subnetted, 4 subnets
O E2   10.1.1.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
O E2   10.1.2.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
O E2   10.1.3.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
O E2   10.1.4.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
 20.0.0.0/30 is subnetted, 3 subnets
C       20.1.1.0 is directly connected, Serial0/1/0
C       20.1.2.0 is directly connected, Serial0/0/0
O       20.1.3.0 [110/128] via 20.1.2.2, 00:27:09, Serial0/0/0
172.16.0.0/24 is subnetted, 12 subnets
O E2   172.16.10.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
O E2   172.16.11.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
O E2   172.16.12.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
C       172.16.20.0 is directly connected, FastEthernet0/0
O       172.16.21.0 [110/65] via 20.1.2.2, 00:27:09, Serial0/0/0
O       172.16.22.0 [110/129] via 20.1.2.2, 00:27:09, Serial0/0/0
O       172.16.23.0 [110/65] via 20.1.2.2, 00:27:09, Serial0/0/0
O       172.16.24.0 [110/129] via 20.1.2.2, 00:27:09, Serial0/0/0
O E2   172.16.30.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
O E2   172.16.31.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
O E2   172.16.32.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
O E2   172.16.33.0 [110/20] via 20.1.1.1, 00:27:09, Serial0/1/0
S*     0.0.0.0/0 [1/0] via 20.1.1.1

```

Copy
Paste


Physical Config **CLI** Attributes

IOS Command Line Interface

```

Hamadullah_R6#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

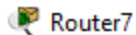
10.0.0.0/30 is subnetted, 4 subnets
O E2   10.1.1.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0
O E2   10.1.2.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0
O E2   10.1.3.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0
O E2   10.1.4.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0
20.0.0.0/30 is subnetted, 3 subnets
O      20.1.1.0 [110/128] via 20.1.2.1, 00:27:54, Serial0/0/0
C      20.1.2.0 is directly connected, Serial0/0/0
C      20.1.3.0 is directly connected, Serial0/0/1
172.16.0.0/24 is subnetted, 12 subnets
O E2   172.16.10.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0
O E2   172.16.11.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0
O E2   172.16.12.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0
O      172.16.20.0 [110/65] via 20.1.2.1, 00:25:44, Serial0/0/0
C      172.16.21.0 is directly connected, FastEthernet0/1
O      172.16.22.0 [110/65] via 20.1.3.2, 00:27:54, Serial0/0/1
C      172.16.23.0 is directly connected, FastEthernet0/0
O      172.16.24.0 [110/65] via 20.1.3.2, 00:27:54, Serial0/0/1
O E2   172.16.30.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0
O E2   172.16.31.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0
O E2   172.16.32.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0
O E2   172.16.33.0 [110/20] via 20.1.2.1, 00:27:44, Serial0/0/0

Hamadullah_R6#
Hamadullah_R6#

```

Copy

Paste



Physical Config **CLI** Attributes

IOS Command Line Interface

```
Hammadullah_R7#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 4 subnets
O E2   10.1.1.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1
O E2   10.1.2.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1
O E2   10.1.3.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1
O E2   10.1.4.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1
20.0.0.0/30 is subnetted, 3 subnets
O      20.1.1.0 [110/192] via 20.1.3.1, 00:28:12, Serial0/0/1
O      20.1.2.0 [110/128] via 20.1.3.1, 00:28:12, Serial0/0/1
C      20.1.3.0 is directly connected, Serial0/0/1
172.16.0.0/24 is subnetted, 12 subnets
O E2   172.16.10.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1
O E2   172.16.11.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1
O E2   172.16.12.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1
O      172.16.20.0 [110/129] via 20.1.3.1, 00:26:07, Serial0/0/1
O      172.16.21.0 [110/65] via 20.1.3.1, 00:28:12, Serial0/0/1
C      172.16.22.0 is directly connected, FastEthernet0/0
O      172.16.23.0 [110/65] via 20.1.3.1, 00:28:12, Serial0/0/1
C      172.16.24.0 is directly connected, FastEthernet0/1
O E2   172.16.30.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1
O E2   172.16.31.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1
O E2   172.16.32.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1
O E2   172.16.33.0 [110/20] via 20.1.3.1, 00:28:12, Serial0/0/1

Hammadullah_R7#
Hammadullah_R7#
```

Copy

Paste

Router8

Physical Config **CLI** Attributes

IOS Command Line Interface

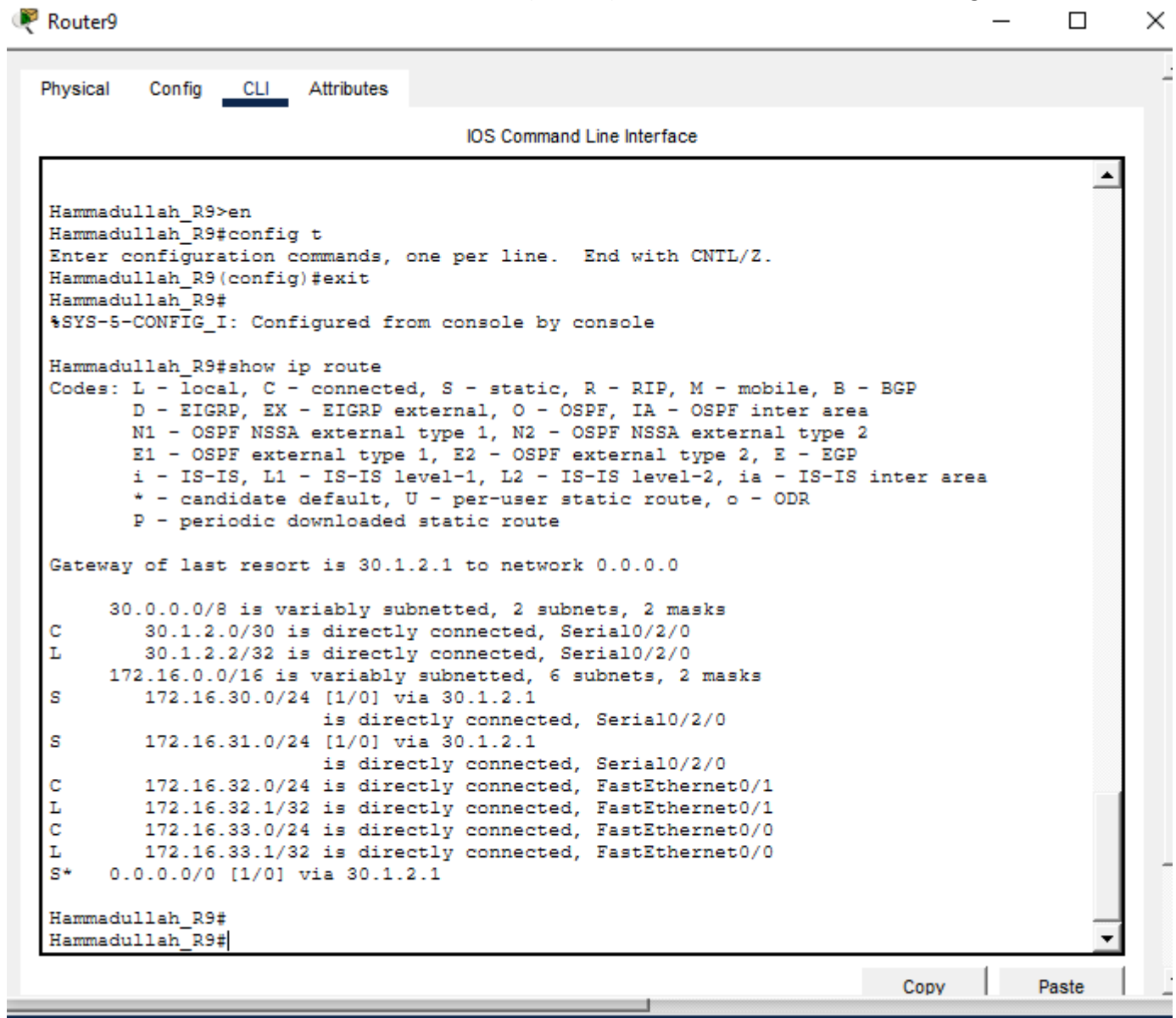
```
Hammadullah_R8>en
Hammadullah_R8#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 30.1.1.1 to network 0.0.0.0

    30.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       30.1.1.0/30 is directly connected, Serial0/2/1
L       30.1.1.2/32 is directly connected, Serial0/2/1
C       30.1.2.0/30 is directly connected, Serial0/2/0
L       30.1.2.1/32 is directly connected, Serial0/2/0
    172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C       172.16.30.0/24 is directly connected, FastEthernet0/0
L       172.16.30.1/32 is directly connected, FastEthernet0/0
C       172.16.31.0/24 is directly connected, FastEthernet0/1
L       172.16.31.1/32 is directly connected, FastEthernet0/1
S       172.16.32.0/24 [1/0] via 30.1.2.2
                        is directly connected, Serial0/2/0
S       172.16.33.0/24 [1/0] via 30.1.2.2
                        is directly connected, Serial0/2/0
S*    0.0.0.0/0 [1/0] via 30.1.1.1

Hammadullah_R8#
Hammadullah_R8#
```

Copy Paste



```
Hammadullah_R9>en
Hammadullah_R9#config t
Enter configuration commands, one per line. End with CNTL/Z.
Hammadullah_R9(config)#exit
Hammadullah_R9#
%SYS-5-CONFIG_I: Configured from console by console

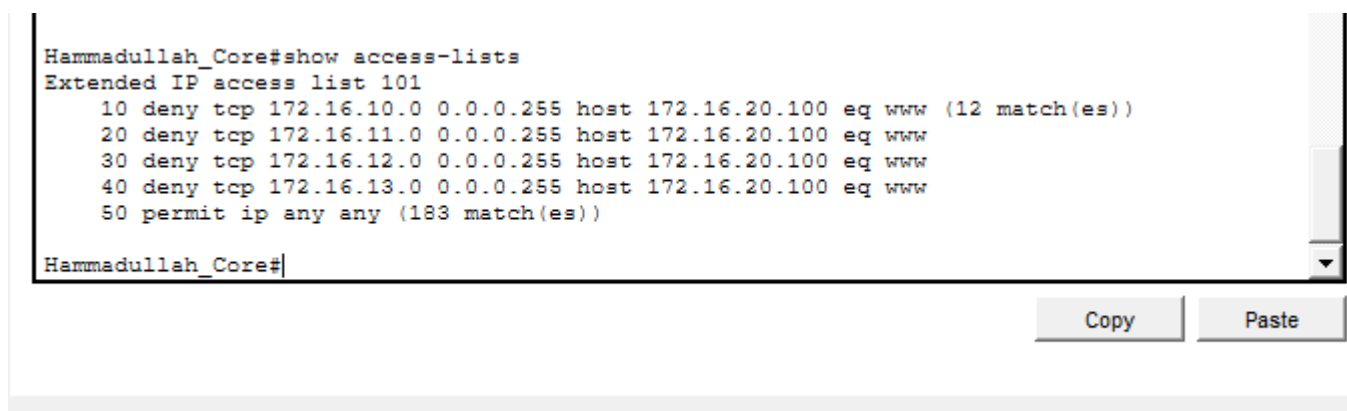
Hammadullah_R9#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 30.1.2.1 to network 0.0.0.0

    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.1.2.0/30 is directly connected, Serial0/2/0
L       30.1.2.2/32 is directly connected, Serial0/2/0
    172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
S       172.16.30.0/24 [1/0] via 30.1.2.1
         is directly connected, Serial0/2/0
S       172.16.31.0/24 [1/0] via 30.1.2.1
         is directly connected, Serial0/2/0
C       172.16.32.0/24 is directly connected, FastEthernet0/1
L       172.16.32.1/32 is directly connected, FastEthernet0/1
C       172.16.33.0/24 is directly connected, FastEthernet0/0
L       172.16.33.1/32 is directly connected, FastEthernet0/0
S*     0.0.0.0/0 [1/0] via 30.1.2.1

Hammadullah_R9#
Hammadullah_R9#
```

ACL CONFIGURATION ON ROUTER4(CORE)



```
Hammadullah_Core#show access-lists
Extended IP access list 101
 10 deny tcp 172.16.10.0 0.0.0.255 host 172.16.20.100 eq www (12 match(es))
 20 deny tcp 172.16.11.0 0.0.0.255 host 172.16.20.100 eq www
 30 deny tcp 172.16.12.0 0.0.0.255 host 172.16.20.100 eq www
 40 deny tcp 172.16.13.0 0.0.0.255 host 172.16.20.100 eq www
 50 permit ip any any (183 match(es))

Hammadullah_Core#
```

WEB SERVER CONFIGURATION

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

HTTP

HTTP ☒ On ☐ Off

HTTPS ☒ On ☐ Off

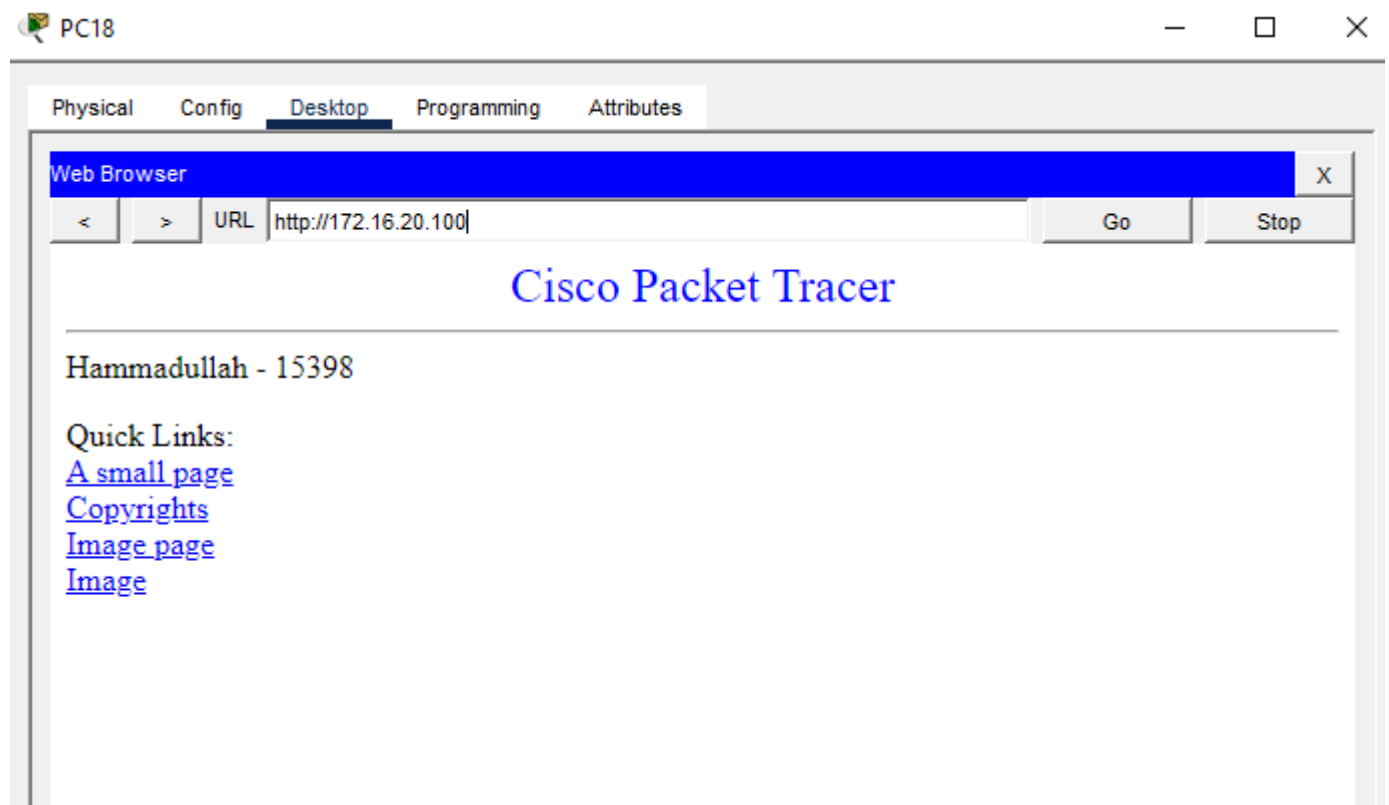
File Manager

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

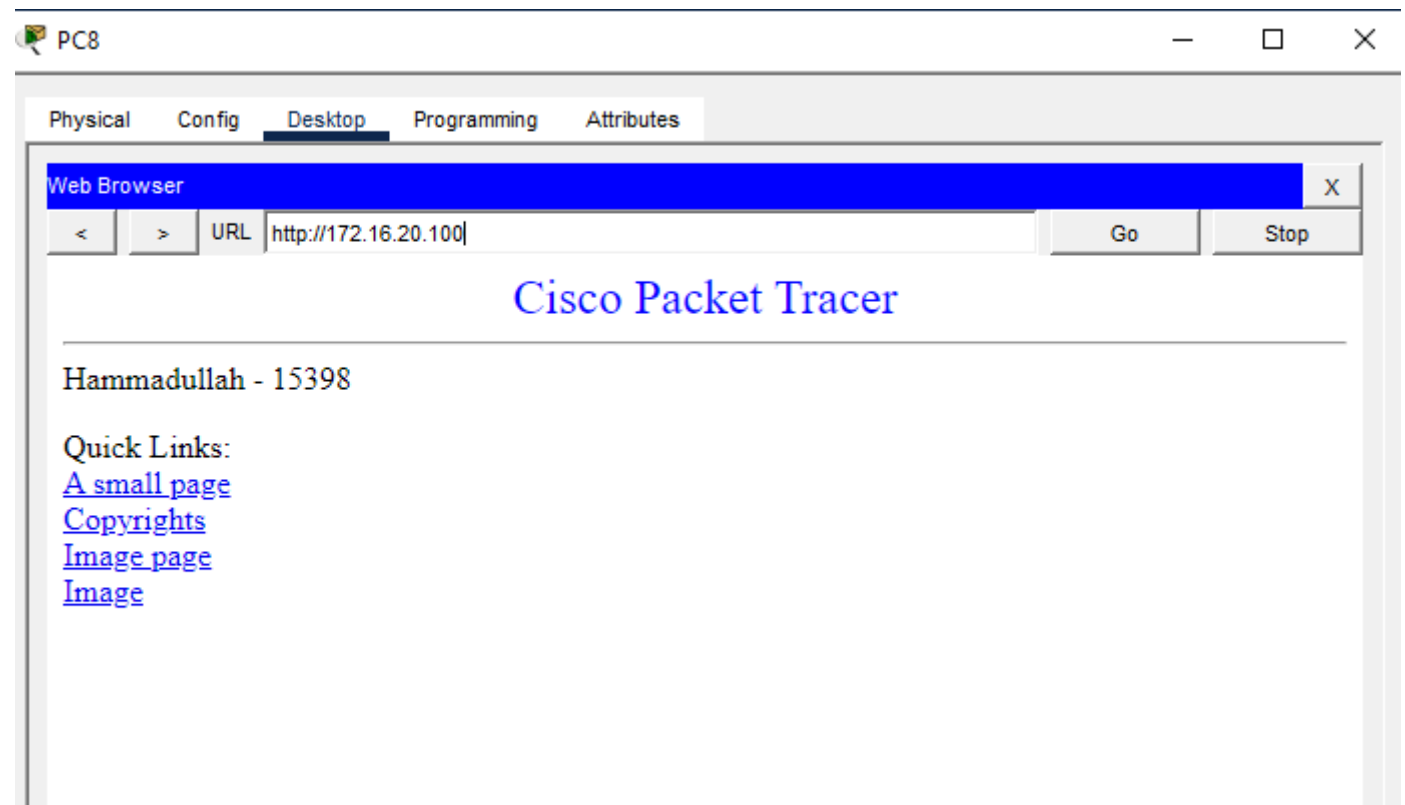
New File Import

Web Access Verification

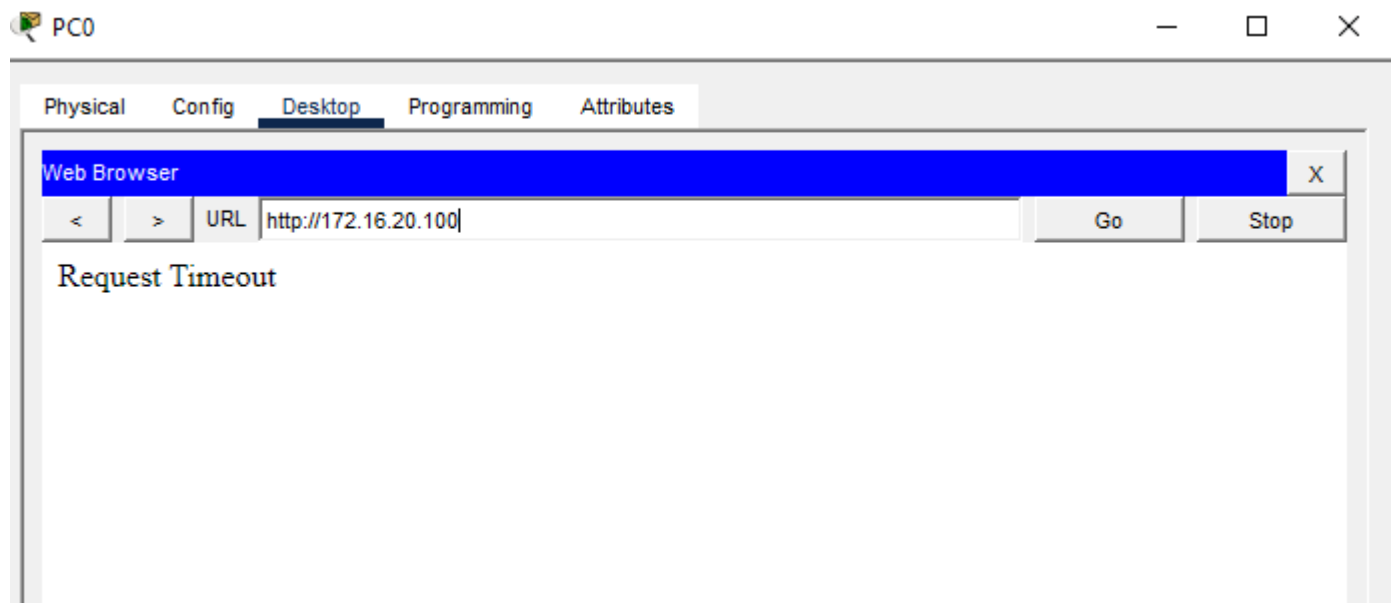
From Static Side



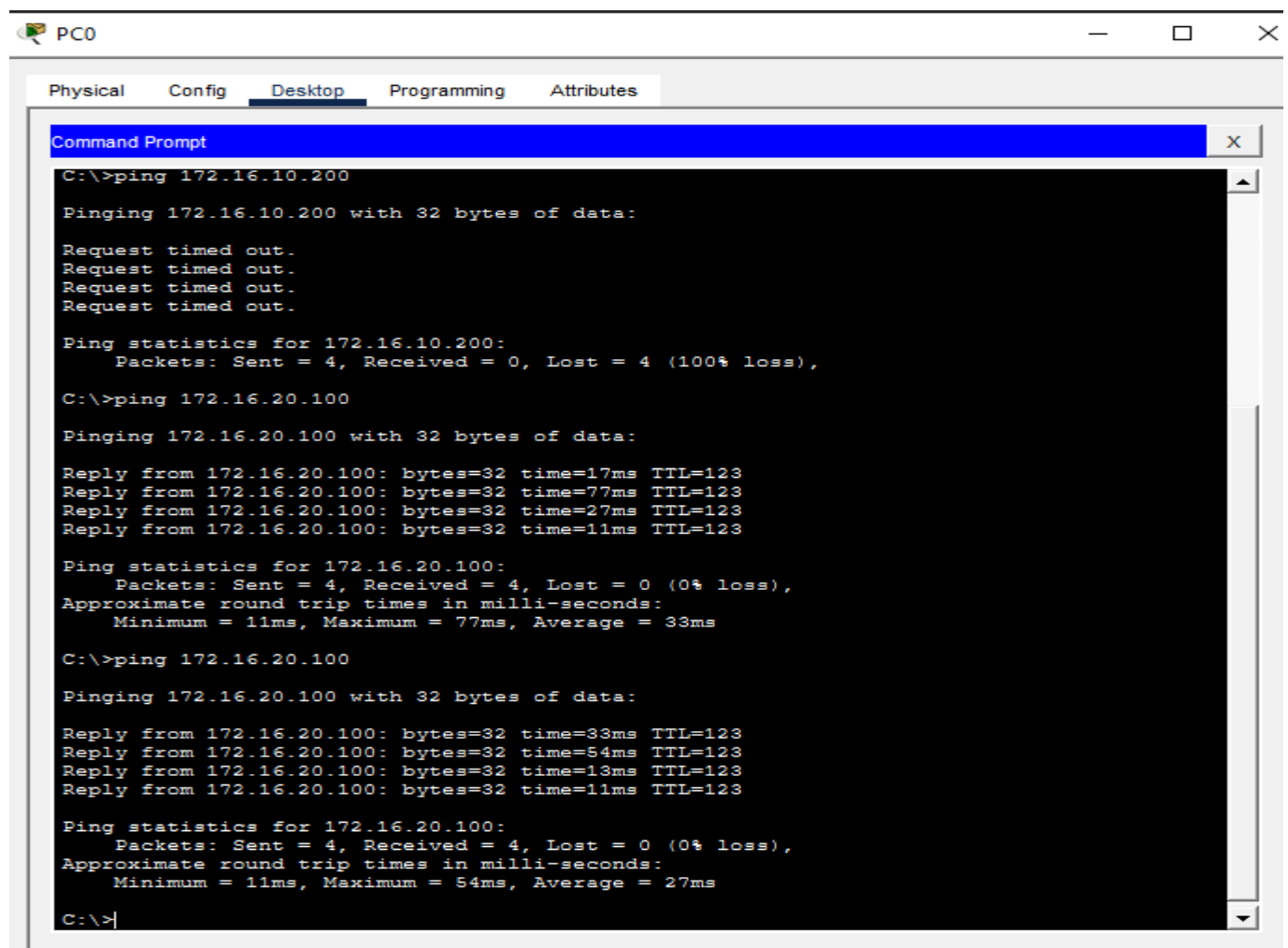
From OSPF Side



From EIGRP SIDE(denied)



From EIGRP Side Ping is Successful



The network was successfully configured with OSPF, EIGRP, and Static routing. Mutual redistribution was performed on the Core Router. An Extended ACL was applied to restrict web access for the EIGRP zone while allowing all other traffic and inter-zone pings.